

WASP-52b

R-0530

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_200818 · created 2026-07-21T08:44:54+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459079.7767 +/- 0.0057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.216 +/- 0.062
Transit depth (Rp/Rs)^2	4.67 +/- 2.69 [%]
Orbital Inclination (inc)	89.82 +/- 2.73
Airmass coefficient 1 (a1)	24.61 +/- 12.4
Airmass coefficient 2 (a2)	-2.98 +/- 0.29
Scatter in the residuals of the lightcurve fit is	41.95 %
Best Comparison Star	#3 - [369, 392]
Optimal Aperture	1.58
Optimal Annulus	3.0
Transit Duration (day)	0.0666 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.216 ± 0.062 vs 0.1646 (deviation 0.83σ)
Duration fit vs published	0.0666 d vs 0.0758 d (deviation 1.34σ)
Mid-time O–C	8.7 min
Depth SNR	3.5
Residual scatter	41.95 %

Light curve

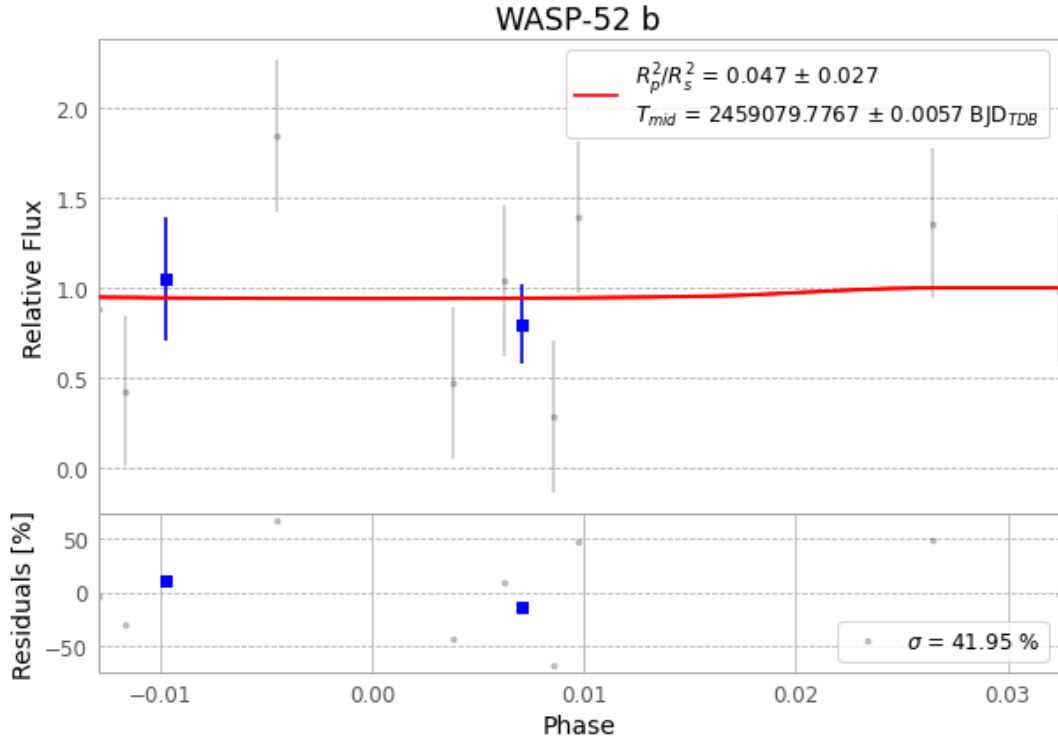


Figure 1 — FinalLightCurve_WASP-52 b_2020-08-17.png (SHA-256 0b8bae9d0ab557fe...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [160, 330], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b216df40be1bdb02...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 72f1d3a

Data provenance

53 FITS frames (35 MB), WASP-52200818055115.FITS → WASP-52200818082715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-200818.log (07ba4fe5834b...), FinalLightCurve_WASP-52 b_2020-08-17.png (0b8bae9d0ab5...), FinalLightCurve_WASP-52 b_2020-08-17.csv (16add175d8d2...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-21 08:44Z.